

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 3 – MCQs

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO3: Articulation (BL4, BL5)	Assessment:	Assessment Tool 3 – MCQs
Weightage:	10% of CIA	Total Marks:	100
CO3 Statement:	Design Divide and Conquer algorithms for Merge Sort, Quick Sort, Binary Search and Matrix Multiplication		
Student Name:	Sasikumar Megha	Reg. No.:	192471038
Section / Batch:	3 rd CSE BS	Date:	17/08/26

Instructions to Students

- Answer all questions carefully. Each question carries 2 marks. Total: 100 marks.
- Analyze each question and select the correct answer.
- Apply concepts correctly to solve problem-based MCQs.
- Manage your time efficiently to complete all 50 questions.

Question Type	Focus Area	Questions
MCQs	Analyze and apply Divide and Conquer Algorithms for Merge Sort, Quick Sort, Binary Search, and Matrix Multiplication.	Q1 -Q50

SECTION A – MCQs

MCQ Link :

(96) ✓

Code of Conduct

I Sasikumar Megha (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: S. Megha

RUBRICS FOR CALCULATION

Online Quiz / MCQ Test

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

Mark Evaluations:

No. of MCQ Questions	Correct Answers	Wrong/Not Answers	Total Marks (100)
50	48	2	96

Faculty In-charge	Course Coordinator	Head of Department
Name: <u>P. Jeyaraj</u>	Signature: _____	Signature: _____
Signature: _____	Signature: _____	Signature: _____
Date: <u>17/8/20</u>	Date: _____	Date: _____

SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Design and Analysis of Algorithms

COURSE CODE: CSA06

COURSE OUTCOME COVERED

CO3: Design Divide and Conquer algorithms for Merge Sort, Quick Sort, Binary Search and Matrix Multiplication (BL4, BL5)

Assessment Tool Weightage: 10%

QUESTIONS: 50 Total Marks: 50

Annexure A

MCQ Test

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Signature of the student: Sasikumar Megha

Annexure A

MCQ Test

Q1. Consider the recurrence relation, which represents a divide-and-conquer algorithm with increasing recursive calls. Analyze how the complexity grows with input size. What is the time complexity?

- A) $O(n \log n)$
- B) $O(n^2)$
- ~~C) $O(n^{\log n})$~~
- D) $O(\log n)$

Q2. Strassen's algorithm improves matrix multiplication by reducing the number of multiplications compared to the conventional method. This makes it more efficient for large matrices. What is its time complexity?

- A) $O(n^3)$
- ~~B) $O(n^2.81)$~~
- C) $O(n \log n)$
- D) $O(n^2)$

Q3. In Quick Sort, the choice of pivot significantly affects performance. If the pivot selected is always the smallest element, analyze the behavior of the algorithm.

What is the worst-case time complexity?

- ~~A) $O(n^2)$~~
- B) $O(n \log n)$
- C) $O(\log n)$
- D) $O(n)$

Q4. Binary Search is an efficient searching technique that repeatedly divides the search space. However, it works only under a specific condition. Identify the required condition.

- A) Randomized data

- ☒ B) Sorted data
- C) Balanced data
- D) Partitioned data

Q5. Merge Sort follows a divide-and-conquer strategy by splitting arrays and merging sorted halves. Identify the correct recurrence relation representing Merge Sort.

- ☒ A) $T(n) = 2T(n/2) + O(n)$
- B) $T(n) = T(n-1) + O(1)$
- C) $T(n) = T(n/2) + O(1)$
- D) $T(n) = 2T(n/2) + O(1)$

Q6. Binary Search reduces the problem size by half in each step. For an array of size 1024, determine the maximum number of comparisons required in the worst case.

- A) 9
- B) 10
- ☒ C) 11
- D) 12

Q7. Strassen's algorithm optimizes matrix multiplication by reducing the number of multiplication operations. Compared to the standard method, how many multiplications are reduced?

- ☒ A) 8 to 7
- B) 7 to 6
- C) 9 to 7
- D) 8 to 6

Q8. Quick Sort performance depends heavily on pivot selection. If the input array is already sorted and the first element is chosen as pivot, analyze the case produced.

- A) Best case
- ☒ B) Worst case

C) Average case

D) None

Q9. Merge Sort and Quick Sort differ in stability when handling duplicate elements. Analyze their behavior and identify the stable algorithm.

☒ A) Merge Sort

B) Quick Sort

C) Both

D) Neither

Q10. Hybrid sorting algorithms combine multiple strategies to improve efficiency.

A common approach switches to Insertion Sort for small subarrays. Why is this optimization used?

A) Reduce recursion depth

☒ B) Improve performance on small inputs

C) Avoid pivot selection

D) Reduce space complexity

Q11. Binary Search uses divide-and-conquer by halving the search space repeatedly. Identify its recurrence relation.

☒ A) $T(n) = T(n/2) + O(1)$

B) $T(n) = T(n-1) + O(1)$

C) $T(n) = 2T(n/2) + O(n)$

D) $T(n) = T(n/2) + O(n)$

Q12. Merge Sort requires additional memory to merge sorted subarrays. Analyze its auxiliary space complexity.

A) $O(1)$

B) $O(\log n)$

☒ C) $O(n)$

D) $O(n^2)$

Q13. Quick Sort is not a stable sorting algorithm due to its partitioning method.

What is the main reason for this behavior?

A) Pivot selection

☒ B) In-place swaps

C) Recursion depth

D) Randomization

Q14. Binary Search tree traversal visits all nodes sequentially. Analyze the time complexity of traversal operations.

~~A) $O(n)$~~

B) $O(\log n)$

C) $O(n \log n)$

D) $O(n^2)$

Q15. Strassen's algorithm is efficient because it reduces the number of multiplication operations. What primary operation is reduced compared to naive multiplication?

A) Additions

~~B) Multiplications~~

C) Memory usage

D) Recursion depth

Q16. Divide and Conquer is a fundamental algorithmic paradigm. Identify the correct sequence of steps involved in this approach.

~~A) Break $\rightarrow$ Solve $\rightarrow$ Combine~~

B) Iterate $\rightarrow$ Optimize $\rightarrow$ Return

C) Choose $\rightarrow$ Compare $\rightarrow$ Select

D) Store $\rightarrow$ Reuse $\rightarrow$ Update

Q17. Binary Search cannot be applied efficiently in certain data structures.

Identify the structure where Binary Search is not suitable.

A) Sorted arrays

~~B) Linked lists~~

C) Balanced trees

D) Hash tables

Q18. Merge Sort divides the array recursively until base cases are reached.

Analyze the height of its recursion tree.

~~A) $\log n$~~

B) n

C) $n \log n$

D) n^2

Q19. Quick Sort achieves best performance when partitions are balanced. Under which condition does this occur?

☒ A) Pivot divides array equally

B) Pivot is smallest

C) Pivot is largest

D) Pivot is random

Q20. Binary Search reduces comparisons logarithmically. For 1000 elements, determine the worst-case number of comparisons.

A) 9

☒ B) 10

C) 11

D) 12

Q21. Strassen's algorithm performs better than naive multiplication only under certain conditions. When is it most efficient?

☒ A) Large input size

B) Small input size

C) $n = 2$

D) $n = 3$

Q22. Merge Sort is considered stable because it preserves the relative order of elements. What property ensures this stability?

☒ A) Preserves order during merging

B) Uses pivot selection

C) Uses recursion

D) Uses partitioning

Q23. Quick Sort average-case performance is efficient due to balanced partitions in most cases. What is its average time complexity?

A) $O(n^2)$

☒ B) $O(n \log n)$

C) $O(\log n)$

D) $O(n)$

Q24. Binary Search computes the middle index during each step. Which formula is commonly used to find the mid value?

☒ A) $(\text{low} + \text{high}) / 2$

☐ B) $(\text{low} + \text{high}) / 3$

☐ C) $(\text{low} + \text{high}) / 4$

☐ D) $(\text{low} + \text{high}) / n$

Q25. Merge Sort is preferred over Quick Sort in certain scenarios such as linked lists. What is the primary reason for this preference?

☒ A) Stability and sequential access

☐ B) Faster pivot selection

☐ C) Less recursion

☐ D) Lower memory usage

Q26. Merge Sort combines smaller subarrays after dividing the original array recursively. This merging process ensures that the final array is sorted efficiently.

What type of arrays are merged during this process?

☐ A) Unsorted arrays

☒ B) Sorted subarrays

☐ C) Random elements

☐ D) Single elements only

Q27. Quick Sort uses a partitioning technique where a pivot element is selected and placed in its correct position. This ensures elements are rearranged around the pivot.

What happens to the pivot after partitioning?

☐ A) Placed randomly

☒ B) Moves to correct sorted position

☐ C) Moves to first index

☐ D) Moves to last index

Q28. Binary Search efficiently finds an element by repeatedly dividing the search space. In some cases, the element may be found immediately.

What is the best-case time complexity of Binary Search?

☐ A) $O(n)$

B) $O(\log n)$

~~C) $O(1)$~~

D) $O(n \log n)$

Q29. Merge Sort divides the input array into smaller parts before sorting and merging them. This division continues until base conditions are reached.

Into how many parts is the array divided at each step?

A) 3 parts

~~B) 2 parts~~

C) n parts

D) $\log n$ parts

Q30. Quick Sort performance depends on how well the pivot divides the array.

Poor pivot selection leads to inefficient execution.

When does Quick Sort exhibit worst-case behavior?

A) Pivot is median

B) Pivot divides equally

~~C) Pivot is smallest or largest~~

D) Input is random

Q31. Binary Search compares the target value with a specific element in the dataset at each step. This comparison reduces the search space efficiently.

Which element is used for comparison?

A) First element

B) Last element

~~C) Middle element~~

D) Random element

Q32. Merge Sort maintains consistent performance regardless of input order due to its divide and conquer approach. This makes it reliable for large datasets.

What is its time complexity in all cases?

A) $O(n^2)$

~~B) $O(n \log n)$~~

C) $O(\log n)$

D) $O(n)$

Q33. Quick Sort is often preferred in practical scenarios due to its in-place nature. It does not require extra memory like Merge Sort.

What is the main advantage of Quick Sort?

- A) Always faster
- ☒ B) Uses less auxiliary space
- C) Stable sorting
- D) No recursion

Q34. Strassen's algorithm optimizes matrix multiplication by reducing the number of multiplication operations. This improves performance for large matrices.

How many multiplications are reduced compared to the standard method?

- A) 6 to 5
- ☒ B) 8 to 7
- C) 9 to 7
- D) 8 to 6

Q35. Binary Search requires a specific condition to work correctly. Without this condition, the algorithm cannot function as expected.

When does Binary Search fail?

- A) Small dataset
- ☒ B) Unsorted data
- C) Large dataset
- D) Numeric data

Q36. Merge Sort is a recursive algorithm that divides the problem into smaller subproblems and solves them independently. This is a key feature of a particular paradigm.

Which paradigm does it follow?

- A) Iterative
- ☒ B) Divide and Conquer
- C) Greedy
- D) Dynamic Programming

Q37. Quick Sort performs partitioning at each recursive step. The partition operation rearranges elements relative to the pivot.

What is the time complexity of partitioning?

- A) $O(1)$
- B) $O(\log n)$
- ~~C) $O(n)$~~
- D) $O(n^2)$

Q38. Binary Search reduces the search space logarithmically with each comparison. This results in fewer comparisons even for large datasets.

What is the worst-case number of comparisons?

- A) n
- ~~B) $\log n$~~
- C) $n \log n$

D) n^2

Q39. Merge Sort continues dividing the array until it reaches its smallest unit. This forms the base case for recursion.

When does this division stop?

- A) Size becomes 2
- ~~B) Size becomes 1~~
- C) Size becomes $n/2$
- D) Array is sorted

Q40. Quick Sort creates recursive calls based on partitioning of the array.

Balanced partitions result in efficient recursion depth.

What is the average recursion depth?

- A) n
- ~~B) $\log n$~~
- C) $n \log n$
- D) 1

Q41. Strassen's algorithm applies a recursive divide and conquer strategy to multiply matrices efficiently. It reduces computational cost.

Which technique does it use?

- A) Greedy
- ~~B) Divide and Conquer~~
- C) Backtracking

D) Brute Force

Q42. Binary Search may fail to find an element even after several comparisons.

This depends on whether the element exists in the dataset.

What is its complexity in unsuccessful search?

A) $O(1)$

~~B) $O(\log n)$~~

C) $O(n)$

D) $O(n \log n)$

Q43. Merge Sort requires extra memory to store temporary arrays during merging. This affects its space efficiency.

Why is it not in-place?

A) Uses recursion

~~B) Needs extra memory~~

C) Uses comparisons

D) Uses pivot

Q44. Quick Sort performance depends heavily on pivot selection strategy. Poor selection can degrade efficiency significantly.

When does Quick Sort become inefficient?

A) Random input

~~B) Poor pivot selection~~

C) Small input

D) Balanced partition

Q45. Binary Search is widely used for efficient searching in large datasets. Its performance depends on data arrangement.

When is Binary Search most effective?

~~A) Sorted data~~

B) Unsorted data

C) Large data

D) Duplicate data

Q46. Merge Sort combines sorted halves to produce a fully sorted array. This combining step is critical to the algorithm.

What is this step called?

A) Partition

~~B) Merge~~

C) Divide

D) Swap

Q47. Quick Sort is considered an in-place algorithm because it minimizes additional memory usage. It modifies the array directly.

Why is it called in-place?

A) No recursion

~~B) Constant extra space~~

C) Uses merging

D) Uses DP

Q48. Strassen's algorithm achieves improved performance by reducing costly operations. This is beneficial for large matrices.

Which operation is minimized?

A) Addition

~~B) Multiplication~~

C) Memory

D) Division

Q49. Binary Search works by dividing the dataset into smaller portions. This ensures faster searching compared to linear methods.

Into how many parts is the data divided?

~~A) Two halves~~

B) Three parts

C) Unequal parts

D) Random parts

Q50. Merge Sort is often used in external sorting where data cannot fit into memory. Its properties make it suitable for such scenarios.

Why is Merge Sort preferred?

A) Faster pivot

~~B) Stability and sequential access~~

C) Less recursion